

A Spurious Minimum for the Logarithmic Fekete Problem on S^2 using Gröbner Bases and Interval Arithmetic via *msolve*

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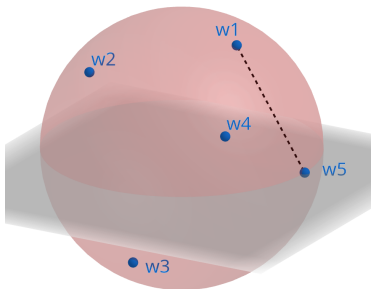
Classifying the Diagonal Blocks

Fekete Problem

Position n points on the sphere $S^d \subseteq \mathbb{R}^{d+1}$, so as to maximize the product of the distances between pairs of points:

$$\max \prod_{1 \leq i < j \leq n} \|w_i - w_j\|,$$

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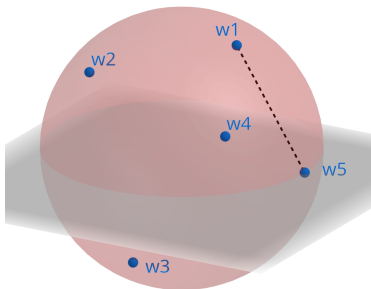


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Equivalently: place the points to minimize the Logarithmic Energy

$$E_{\log}(w) := - \sum_{1 \leq i < j \leq n} \log (\|w_i - w_j\|^2), \quad w_i \in S^d \subseteq \mathbb{R}^{d+1}.$$

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Smale's 7th problem for the next Century (1998) [7]

Give an “efficient” algorithm that, for each number of points n , finds $\hat{w} \in (S^2)^n$ with optimal value up to a logarithmic factor:

$$E_{\log}(\hat{w}) - E_{\log}^* \leq c \log(n), \quad \text{for a universal constant } c.$$

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$$\text{in } S^2: \quad E_{\log}^* = \left(\frac{1}{2} - \log 2 \right) \frac{n^2}{2} - \frac{n \log n}{4} + Cn + o(n), \quad C = ?$$

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- ▶ Cannot have local **maxima**.
- ▶ What about **spurious minima**? (i.e. non global local minima)

Logarithmic Fekete problem - Spurious Minima

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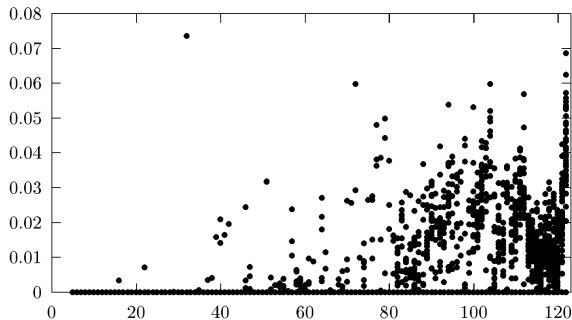
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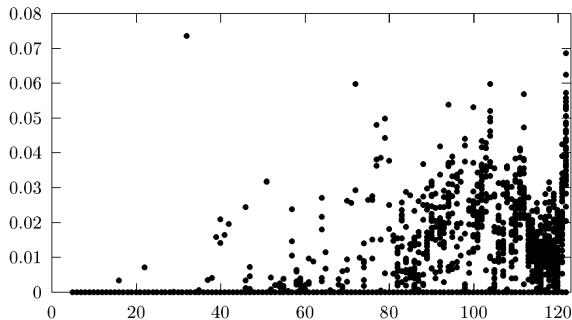
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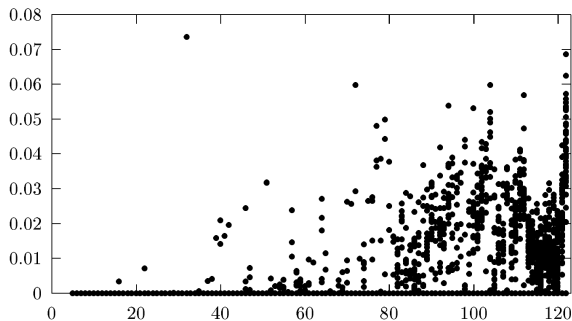


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- ▶ Rigorously: no spurious minimum on S^2 is known
- ▶ We give rigorous proof of spurious minimum on S^2 , for $n = 16$

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- ▶ $n = 16$ points: smallest with evidence of spurious minima.
- ▶ Configuration 4:4:4:4: four squares, at different heights $\pm h_1, \pm h_2$, with 45 degrees between consecutive squares.

Figure: Squares on the sphere S^2

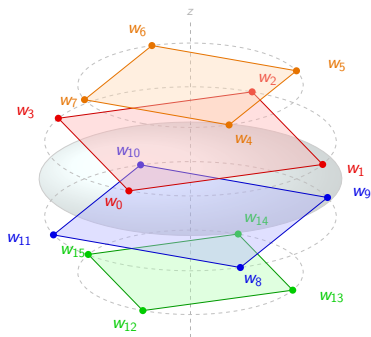
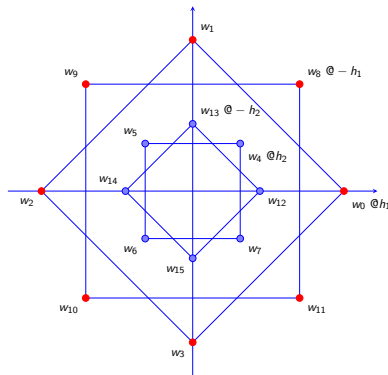


Figure: Configuration from above



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Critical configurations (null gradient condition $\frac{\partial L}{\partial w_i}(w) = \vec{0}$):

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Critical conditions at configuration 4:4:4:4, in terms of the heights:

$$\begin{aligned} @w_0 : \quad & \frac{h_1 (h_1 + h_2) p(h_1, h_2)}{\sqrt{(1-h_1)(1+h_1)} q(h_1, h_2)} = 0, \quad \frac{(h_1 + h_2) p(h_1, h_2)}{q(h_1, h_2)} = 0, \\ @w_4 : \quad & \frac{h_2 (h_1 + h_2) p(h_2, h_1)}{\sqrt{(1-h_2)(1+h_2)} q(h_2, h_1)} = 0, \quad \frac{(h_1 + h_2) p(h_2, h_1)}{q(h_2, h_1)} = 0; \end{aligned}$$

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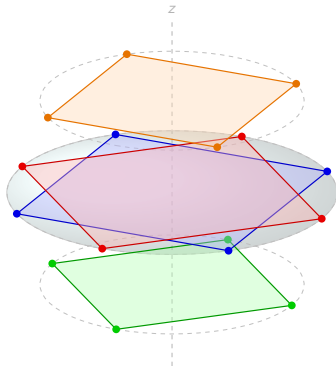
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$$p(h_1, h_2) := 15h_1^9 h_2^3 + 15h_1^9 h_2 + 13h_1^8 h_2^4 - 26h_1^8 h_2^2 + 13h_1^8 + 44h_1^7 h_2^3 + 44h_1^7 h_2 + 65h_1^6 h_2^4 - 130h_1^6 h_2^2 + 65h_1^6 - 134h_1^5 h_2^3 - 134h_1^5 h_2 - 3h_1^4 h_2^4 + 6h_1^4 h_2^2 - 3h_1^4 + 76h_1^3 h_2^3 + 76h_1^3 h_2 - 25h_1^2 h_2^4 + 50h_1^2 h_2^2 - 25h_1^2 - h_1 h_2^3 - h_1 h_2 - 2h_2^4 + 4h_2^2 - 2$$

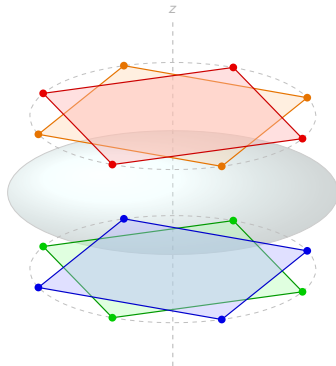
$$q(h_1, h_2) := (h_1 + h_2)^2 (h_1 h_2 + 1) (h_1^4 + 6h_1^2 + 1) (h_1^2 h_2^2 + h_1^2 - 4h_1 h_2 + h_2^2 + 1).$$

Ideal of Critical Heights - Exclude “degenerate” cases

$$h_1 = 0, h_2 \neq 0$$

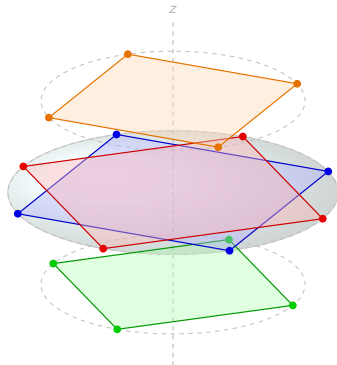


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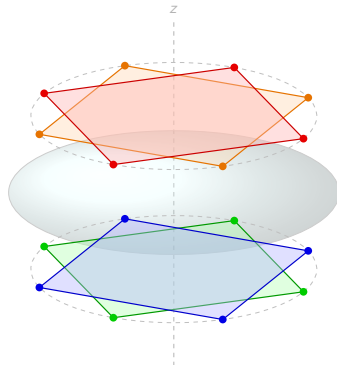


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Ideal of critical heights (with auxiliary variables s_i):

$$p(h_1, h_2) = 0, \quad p(h_2, h_1) = 0, \quad s_1 q(h_1, h_2) = 1, \quad s_2 q(h_2, h_1) = 1,$$

$$s_3(h_1 - h_2) = 1, \quad s_4 h_1 h_2 (h_1 + h_2) (1 - h_1^2) (1 - h_2^2) - 1 = 0.$$

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$$\{h_1 \in [a, b], \quad h_2 \in [c, d]\}, \quad \{h_1 \in [c, d], \quad h_2 \in [a, b]\}.$$

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$$a = -4388364404593850244675593032723597202310835/2^{142} \simeq -0.787,$$

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- ▶ By symmetry: system has only one “kind” of real solution.
- ▶ Work modulo the ideal of critical heights, to restrict results (about real critical configurations) to configuration 4:4:4:4.

Hessian of the Lagrangian at 4:4:4:4

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- ▶ We will see that Hessian is positive definite in **all** directions of $\mathbb{R}^{48}$ (except the ones corresponding to rotations of the sphere)

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Value	0	0.175	0.519	0.905	1.55	2.70	3.27	4.43	4.44
Alg. mult.	3	2	2	2	1	1	2	1	2

Value	5.48	6.15	6.28	7.18	8.81	9.05	9.19	10.6	10.6
Alg. mult.	1	2	2	2	2	1	2	2	1

Value	11.7	12.3	13.5	14.1	14.5	14.8	15	30	
Alg. mult.	2	1	1	2	2	2	6	1	

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Value	5.48	6.15	6.28	7.18	8.81	9.05	9.19	10.6	10.6
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Value	11.7	12.3	13.5	14.1	14.5	14.8	15	30	
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- ▶ But: Hessian is dense symbolic matrix, with $48^2 = 2304$ entries which are rational functions.
- ▶ **How?** Exploit symmetries of configuration 4:4:4:4 to block diagonalize the Hessian and classify each block separately.

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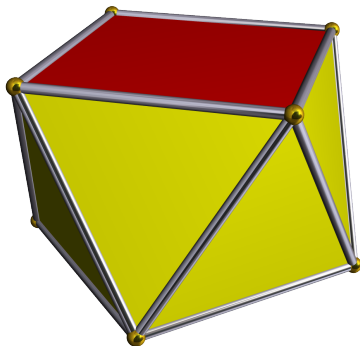


Figure: Square Antiprism (from Wikipedia).

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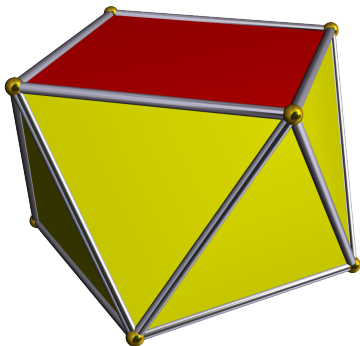


Figure: Square Antiprism (from Wikipedia).

- ▶ Group of symmetries of square antiprism is D_{4d} , isomorphic to D_8 (symmetries of regular polygon of 8 vertices in the plane)

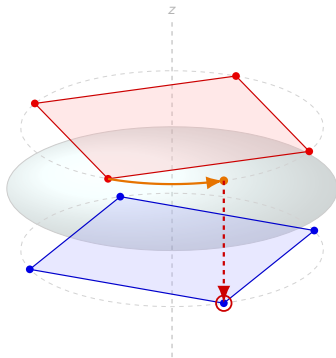
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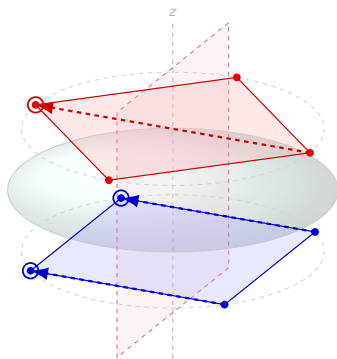
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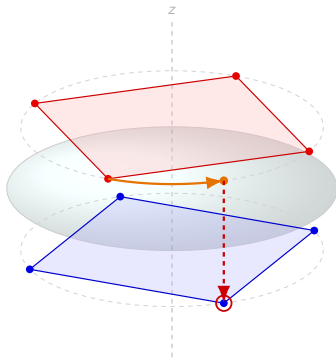
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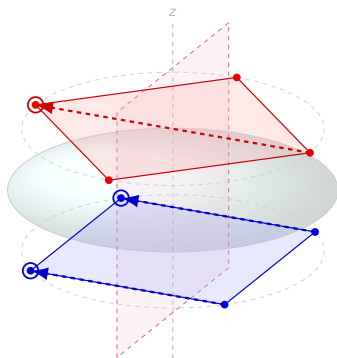
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In analogy with D_8 , with $S_z R_{45}$ playing the role of the rotation in the plane, and S_y of a reflection, we can describe D_{4d} as:

$$D_{4d} = (S_z R_{45}, S_y), \quad (S_z R_{45})^8 = I_3, \quad S_y^2 = I_3, \quad (S_y (S_z R_{45}))^2 = I_3.$$

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1. $\rho : G \rightarrow GL(V)$ associates to each $g \in G$ an invertible linear transformation ($N \times N$ matrix) $\rho(g) : V \rightarrow V$.
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- ▶ By Schur’s Lemma: consider a linear transformation (matrix) $H : V \rightarrow V$ that commutes with a representation (ρ, V) :

$$H\rho(g) = \rho(g)H, \quad \forall g \in G.$$

H is block diagonalized in a basis of the irreducible representations that “build” ρ .

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Theorem

The Hessian of the Lagrangian, evaluated at configuration 4:4:4:4, commutes with our representation $(\rho, \mathbb{R}^{48})$.

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- ▶ Project a basis of $\mathbb{R}^{48}$ on each of these irreps.

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$$\rho(S_z R_{45}) = \left(\begin{array}{c|c|c|c} 0 & 0 & C & 0 \\ 0 & 0 & 0 & D \\ D & 0 & 0 & 0 \\ 0 & C & 0 & 0 \end{array} \right), \quad C := \left(\begin{array}{c|c|c|c} 0 & 0 & 0 & M \\ M & 0 & 0 & 0 \\ 0 & M & 0 & 0 \\ 0 & 0 & M & 0 \end{array} \right), \quad D := \left(\begin{array}{c|c|c|c} M & 0 & 0 & 0 \\ 0 & M & 0 & 0 \\ 0 & 0 & M & 0 \\ 0 & 0 & 0 & M \end{array} \right).$$

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$$\rho\left((S_z R_{45})^k\right) = \rho(S_z R_{45})^k, \quad \rho\left((S_z R_{45})^k S_y\right) = \rho(S_z R_{45})^k \rho(S_y), \quad \forall k.$$

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- ▶ Traces are calculated exactly using *SymPy*, and given by:

$$\chi(e) = 48, \quad \chi(\rho(r^k s)) = \begin{cases} 4, & \forall k = 0, 2, 4, 6 \\ 0, & \forall k = 1, 3, 5, 7 \end{cases},$$

$$\chi(\rho(r^k)) = 0, \quad \forall k \in \{1, \dots, 7\}.$$

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irrep.	irrep. copies $\langle \chi, \chi_i \rangle$	irrep. dimension	copies $\times$ dimension
ρ_1	$3 + 1 = 4$	1	4
ρ_2	$3 + 1 = 4$	1	4
ρ_3	$3 - 1 = 2$	1	2
ρ_4	$3 - 1 = 2$	1	2
ρ_5	$3 \times 2 + 0 = 6$	2	12
ρ_6	$3 \times 2 + 0 = 6$	2	12
ρ_7	$3 \times 2 + 0 = 6$	2	12
		Total dimension	48

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6. For each t : verify that the lower bound of the interval is positive (except three of them).

A detailed example for Block 4, of size 2×2

```
R = QQ[s1, s2, s3, u0, u1, A00, A01, A11, h0, h1, t];

I = initialize critical heights ideal.

-- define sqrt{1-h0^2}, sqrt{1-h1^2} for some Aij
I = I + ideal( u0^2 - ( 1-h0^2 ), u1^2 - ( 1-h1^2 ) );

-- add characteristic polynomial
A = matrix{ {A00, A01}, {A01, A11} };
I = I + ideal( det(A - t * matrix{{1,0},{0,1}} ) );

-- Aij are rational functions in the other variables
I = I + ideal(
    A00 * (h0^18*h1^6 + ... ) - (... - 24*h1^2 + 2),
    A01 * (h0^8*h1^6 + ... ) - (... + 2*u0*u1),
    A11 * (h0^8*h1^16 + ... ) - (... - 2*h1^2 + 2)
);
```

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- ▶ Projected ideal has variables h_0, h_1, t only.
- ▶ Reduced Gröbner basis of this ideal has 16 polynomials, with “big” coefficients:

$371292491088505461729459020333037295820800t^6 -$
 $162141499362359817045945195361214200795449560003625h_0^4t - \dots +$
 $118210982313945344981292610227750651610274227426413879,$

$947174722164554749309844439625095142400h_0h_1t^4 +$
 $129107873997955695356479196241141905858467125h_0^4t + \dots +$
 $4451044050262360998981074020917432846944434053t -$
 $24236909582057465464703668056872429827406416527,$

$\vdots$

$58273444820670848834492382515997064425000h_0^2h_1^5 -$
 $19872276148503933316980431005986760058625h_0^5t - \dots +$
 $8317535443862061136887544113047987541013101h_1$

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- ▶ Projected ideal has exactly 8 real solutions (h_0, h_1, t) ; four solutions for each eigenvalue t of the 2×2 matrix.
- ▶ Remember that ideal of critical heights has exactly 4 real solutions (h_0, h_1) , all of the “kind” 4:4:4:4.
- ▶ Coordinates of one of the eight real solutions are bounded as:

$$h_0 \in [66587440247399979866571509524517571051/2^{128}, \\ 16646860061849994966642877381129392763/2^{126}] \simeq [.196, .196],$$

$$h_1 \in [267844507116323867472875551313696118305/2^{128}, \\ 133922253558161933736437775656848059153/2^{127}] \simeq [.787, .787],$$

$$t \in [899504581213468778226187660896150819527/2^{126}, \\ 3598018324853875112904750643584603278109/2^{128}] \simeq [10.57, 10.57].$$

Merci de votre attention

Code and Presentation

<http://www.github.com/matiasvd>



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Hessian of the Lagrangian

1. For n points, the Hessian of the Lagrangian is a $3n \times 3n$ matrix H_L , made by blocks of 3×3 matrices, each given by:

$$(H_L)_{ii} := \frac{\partial L}{\partial w_i^2} = \sum_{j=1, j \neq i} J_f(w_i - w_j) + (n-1)\text{Id}_3, \quad \forall i.$$

$$(H_L)_{ij} := \frac{\partial L}{\partial w_i w_j} = -J_f(w_i - w_j), \quad \forall i \neq j;$$

where we define an auxiliary function $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, with Jacobian:

$$J_f(w) := \frac{-2}{\|w\|_2^4} \begin{pmatrix} -x^2 + y^2 + z^2 & -2xy & -2xz \\ -2xy & x^2 - y^2 + z^2 & -2yz \\ -2xz & -2yz & x^2 + y^2 - z^2 \end{pmatrix}.$$

2. Entries of H_L are rational functions (quotients of polynomials)

Characters of irreps. of D_{4d} and of our representation

Denote $r := S_z R_{45}$, $s := S_y$. Equivalence classes of D_{4d} are:

$$\{e\}, \{r, r^7\}, \{r^2, r^6\}, \{r^3, r^5\}, \{r^4\}, \{s, r^2s, r^4s, r^6s\}, \{rs, r^3s, r^5s, r^7s\}.$$

Table: Characters of the irreducible representations of D_{4d} .

irrep. / class	[e]	[r]	[r ²]	[r ³]	[r ⁴]	[s]	[rs]
ρ_1	1	1	1	1	1	1	1
ρ_2	1	-1	1	-1	1	1	-1
ρ_3	1	1	1	1	1	-1	-1
ρ_4	1	-1	1	-1	1	-1	1
ρ_5	2	$\sqrt{2}$	0	$-\sqrt{2}$	-2	0	0
ρ_6	2	0	-2	0	2	0	0
ρ_7	2	$-\sqrt{2}$	0	$\sqrt{2}$	-2	0	0

Table: Character χ of our representation ρ . $r := S_z R_{45}$, $s := S_y$

class	[e]	[r]	[r ²]	[r ³]	[r ⁴]	[s]	[rs]
χ	48	0	0	0	0	4	0

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